



Electric vehicle's electricity consumption on a road with different slope



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HIGHLIGHTS

- An extended car-following model is proposed.
- An electricity consumption model is proposed.
- The electric vehicle's electricity consumption is explored on uphill.
- The electric vehicle's electricity consumption is studied on downhill.

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ABSTRACT

In this paper, we propose an extended car-following model and an electricity consumption model to study the effects of the road's slope on the electric vehicle's electricity consumption. The numerical results show that each electric vehicle's electricity consumption increases with the uphill's tilt angle and decreases with the downhill's tilt angle. In addition, each electric vehicle's electricity consumption increases with the uphill's (downhill's) length under a certain tilt angle.

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1. Introduction

To date, traffic problems (congestions, accidents, energy consumption) have turned very serious, so researchers have developed many models to explore various complex traffic phenomena [1,2]. Roughly speaking, the models can be divided into three types: the first ones are to reduce traffic accident [3,4]; the second ones are to improve the traffic flow stability or to optimize the traffic flow [5–16]; the third ones are to develop a car-following model to explore the driving behavior [17–23]. Recently, traffic pollution has been serious, so researchers have proposed many models to study the vehicle's exhaust emissions [24–29], but traffic pollution always exists in the existing traffic systems since the traditional vehicles produce exhaust emissions. To reduce the traffic pollution, electric vehicles have been used to replace the traditional vehicles and been an important traffic tool [30,31], which has attracted researchers to propose many models to study electric vehicles (including the electric vehicle's electricity consumption [32,33] and the electric vehicle's battery life [34–36]).

The models [30–36] can describe the electric vehicle's traffic, but they cannot be used to explore the influences of the road's slope on the electric vehicle's electricity consumption. In fact, each road has its slope and the road's slope often influences the driving behavior. Thus, researchers proposed some models to explore the influences of the road's slope on the driving behavior [37–40] and Zhu [41] further studied the vehicle's energy consumption under the model [38], but they

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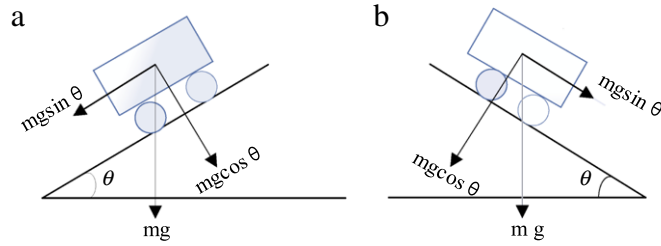


Fig. 1. Illustrations of the electric vehicle's force on uphill and downhill with (a) and (b) respectively.

do not further study the effects of the road's slope on the electric vehicle's electricity consumption. In this paper, we extend the car-following model [37] to investigate the effects of the road's slope on the electric vehicle's electricity consumption under different traffic states.

2. The relation model

In this paper, we suppose that electric vehicles move on a single lane with road slope. The gravitational force affects each vehicle in the traffic system, so the road's slope should be explicitly considered when we study the driving behavior. Fig. 1 displays the impacts of the gravitational force on the vehicle's driving behavior on the uphill/downhill road, where θ is the road tilt angle, g is the gravitational acceleration and m is the electric vehicle's mass. In fact, $mg \sin \theta$ will affect the driving behavior except its braking process, i.e., $mg \sin \theta$ does not influence the driving behavior during the braking process because this term is offset by the braking force [37]. To study the driving behavior on the road with different slopes, Zhu and Yu [37] extended the OV (optimal velocity) model [42] and proposed a new car-following model.

$$\frac{d^2 x_n(t)}{dt^2} = a \left\{ V(\Delta x_n) - \frac{dx_n(t)}{dt} \right\}, \quad (1)$$

where x_n is the n th vehicle's position, Δx_n is the n th vehicle's headway, a is the driver's reaction coefficient, $V(\Delta x_n)$ is the optimal velocity. In addition, $V(\Delta x_n)$ on the uphill can be defined as follows:

$$V(\Delta x_n) = \frac{(v_{f,\max} - v_{g,u,\max}) [\tanh(\Delta x_n - x_{c,u}) + \tanh(x_{c,u})]}{2}, \quad (2)$$

and $V(\Delta x_n)$ on the downhill can be defined as follows:

$$V(\Delta x_n) = \frac{(v_{f,\max} + v_{g,d,\max}) [\tanh(\Delta x_n - x_{c,d}) + \tanh(x_{c,d})]}{2}, \quad (3)$$

where $x_{c,u}$, $x_{c,d}$ are respectively the vehicle's safety distances on the uphill and downhill, $v_{g,u,\max}$, $v_{g,d,\max}$ are respectively the adjustment terms resulted by the uphill and downhill, $v_{f,\max}$ is the maximal speed without the road's slope.

$v_{g,u,\max} = v_{g,d,\max} = mg \cdot \sin \theta$ in the model [37], but they are unreasonable since the dimension of mg is Newton and the dimension of speed is m/s. Therefore, we should here redefine $v_{g,u,\max}$, $v_{g,d,\max}$. For simplicity, we define them as follows:

$$v_{g,u,\max} = v_{g,d,\max} = v_{\text{slope}} \cdot \sin \theta, \quad (4)$$

where v_{slope} is the adjustment term resulted by the road's slope.

Eq. (4) shows that the terms $v_{g,u,\max}$, $v_{g,d,\max}$ are proportional to the road slope, i.e., $v_{g,u,\max}$, $v_{g,d,\max}$ increase with the slope. For simplicity, we define the related parameters as follows:

$$v_{f,\max} = 10, \quad v_{\text{slope}} = 1, \quad v_{g,u,\max} = v_{g,d,\max} = \sin \theta. \quad (5)$$

To emphasize the impacts of the road's slope on the vehicle's driving behavior, we should study the effects of the road's slope on the vehicle's safe distance. In fact, the vehicle's safe distance decreases with the slope of uphill and increases with the slope of downhill. For simplicity, we here define the safe distance as follows [37]:

$$x_{c,u} = x_c (1 - \alpha \sin \theta), \quad x_{c,d} = x_c (1 + \beta \sin \theta), \quad (6)$$

where x_c is the safety distance on the road without slope, in this paper we define $x_c = 20$ m. For simplicity, we set $\alpha = \beta = 1$ and define $V_0(\Delta x_n)$ as follows [37]:

$$V_0(\Delta x_n) = \tanh(\Delta x_n - x(\theta)) + \tanh(x(\theta)), \quad (7)$$

where $x(\theta)$ is defined by $x_{c,u}$ or $x_{c,d}$ that is defined in Eq. (6).

Based on the above discussions, we have [37]

$$V(\Delta x_n) = \left(\frac{2 \pm \sin \theta}{2} \right) V_0(\Delta x_n). \quad (8)$$

Thus, Eq. (1) can be rewritten as follows [37]:

$$\frac{d^2 x_n(t)}{dt^2} = a \left\{ \left(\frac{2 \pm \sin \theta}{2} \right) V_0(\Delta x_n) - \frac{dx_n(t)}{dt} \right\} \quad (9)$$

where $v_n(t) = \frac{dx_n(t)}{dt}$.

The above model can be used to explore the effects of the road's slope on the traditional vehicle's movement behavior on the road with different slopes, but it cannot be used to directly study the influences of the road's slope on the electric vehicle's electricity consumption. In order to explore the electric vehicle's electricity consumption, Mehrdad [43] proposed a model to explore the output power of the electric vehicle's battery, where the control equation can be formulated as follows:

$$P_{b-out} = \frac{v}{\eta_{te}\eta_m} \left(\delta m \frac{dv}{dt} + mg(f + i) + \frac{\rho C_D A}{2} v^2 \right), \quad (10)$$

where P_{b-out} is the output power of the electric vehicle's battery, δ is the coefficient that is related to the electric vehicle's mass, m is the electric vehicle's mass, f is the rolling resistance coefficient, i is the grade that considers the grading resistance, C_D is the aerodynamic drag coefficient, ρ is the air density, A is the electric vehicle's frontal area, η_{te} is the electric vehicle's transmission efficiency, η_m is the motor efficiency.

During the braking process, the traditional vehicle's braking energy is wasted while the electric vehicle's braking energy can partly be recovered and restored into the battery. Thus, based on Eq. (10), the electric vehicle's regenerative braking power can be formulated as follows:

$$P_{b-in} = kv\eta_{te}\eta_m \left(\delta m \frac{dv}{dt} + mg(f + i) + \frac{\rho C_D A}{2} v^2 \right), \quad (11)$$

where k ($0 < k < 1$) is the percentage of the total braking energy that can be recovered by the motor (called the regenerative braking factor) and the regenerative braking factor k is a function of the property of the electric vehicle's brake, the battery property and other related factors.

In this paper, we do not employ Eqs. (10) and (11) to study the electric vehicle's electricity consumption because they consider too many factors. Eq. (10) shows that some of the electric vehicle's electricity is consumed by its resistance, so we should here apply the parameter η_e to substitute the parameter η_m , where η_e is the driving efficiency of the electric vehicle's battery and $\eta_e < \eta_m$. In addition, Eq. (11) does not consider the electricity consumed by other accessories (e.g., the electric power steering) and the electricity consumption may be relatively high, so we should consider the effects of the accessories on the electric vehicle's electricity consumption. Thus, we here rewrite Eq. (10) as follows:

$$P_{total} = \frac{v}{\eta_{te}\eta_e} \left(\delta m \frac{dv}{dt} + mg(f + i) + \frac{\rho C_D A}{2} v^2 \right) + P_{accessory}, \quad (12)$$

and Eq. (11) can be rewritten as follows:

$$P_{total} = kv\eta_{te}\eta_m \left(\delta m \frac{dv}{dt} + mg(f + i) + \frac{\rho C_D A}{2} v^2 \right) + P_{accessory}, \quad (13)$$

where P_{total} is the total power of the electric vehicle's battery.

Based on the relationship between the power and energy, the electric vehicle's electricity consumption can be defined as follows:

$$W = P_{total} \cdot t, \quad (14)$$

where W is the electric vehicle's electricity consumption.

In fact, the parameter k is very complex and related to many factors (e.g., speed, acceleration, etc.), but it is directly related to the regenerative braking force (see Fig. 2) [44]. Thus, based on Fig. 1, we here define k as follows:

$$k = \begin{cases} 0.5 * \frac{v}{5} & v < 5 \text{ m/s} \\ 0.5 + 0.3 \frac{v-5}{20} & v \geq 5 \text{ m/s.} \end{cases} \quad (15)$$

And other related parameters' values are shown in Table 1.

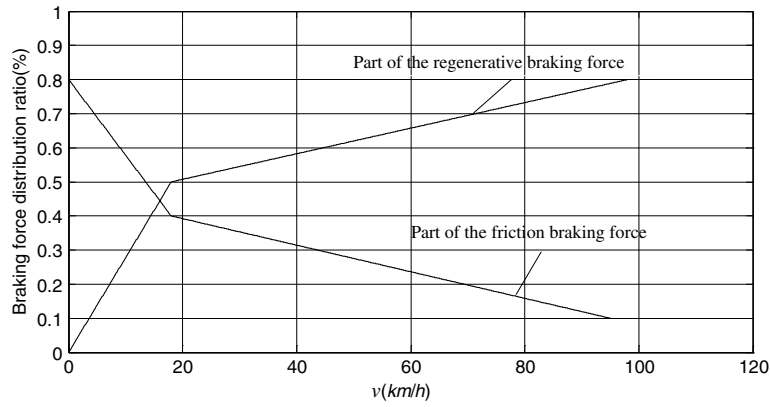


Fig. 2. The distribution of the braking force in ADVISOR [16].

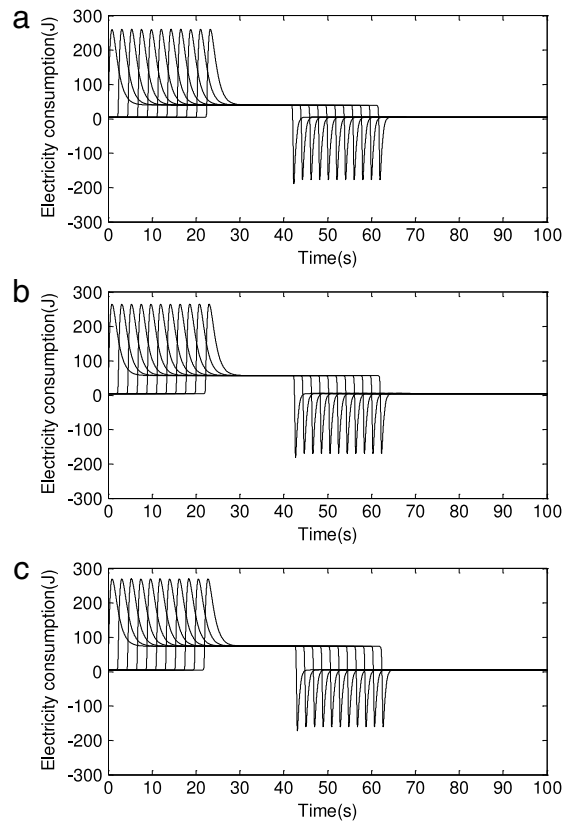


Fig. 3. The electric vehicle's electricity consumption on uphill, where the tilt angles in (a)–(c) are respectively 1° , 2° , 3° .

3. Numerical simulation

In this section, we study the influences of the road's slope on the electric vehicle's electricity consumption. First, we study the effects of the road's slope on the electric vehicle's electricity consumption during the starting and braking processes. The initial conditions are defined as follows:

11 electric vehicles are uniformly distributed on a road with different slope; where the first vehicle is the leading one and the last vehicle lies at the origin; except the leading vehicle, the other 10 vehicles' headways are 7.4 m; all the vehicles are still when $t < 0$ but will immediately start when $t = 0$; all the vehicles should stop when the first vehicle reaches the 500 m since there is an obstacle.

In addition, other parameters are defined in Eqs. (5) and (6).

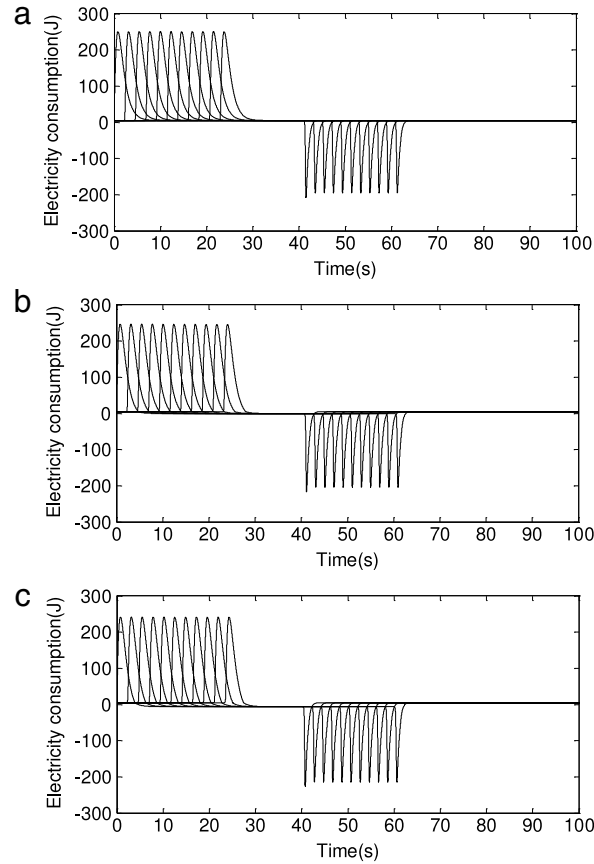


Fig. 4. The electric vehicle's electricity consumption on downhill, where the tilt angles in (a)–(c) are respectively 1° , 2° , 3° .

Table 1

The values of the electric vehicle's related parameters.

δ	1.1
m	1500
f	0.015
C_D	0.3
A	1.8
ρ	1.2
η_{te}	0.85
η_e	0.85
$P_{accessory}$	1000
U	300
I	100

Thus, we obtain each electric vehicle's electricity consumption during the starting and braking processes on uphill and downhill (see Figs. 3 and 4).¹ From the two figures, we can conclude the following findings:

(1) each electric vehicle's electricity consumption can be divided into three prominent stages. In the first stage, each electric vehicle's electricity consumption sharply increases because its speed increases. In the second stage, each electric vehicle's electricity consumption is constant since its speed has been stable. In the third stage, each electric vehicle's electricity consumption is negative since each electric vehicle regenerates electricity during the process. Each electric vehicle's electricity consumption drops zero when it completely stops and each electric vehicle's total electricity consumption during the first stage and second stage is larger than the one during the third stage (see Figs. 3 and 4);

(2) with the increase of the uphill's tilt angle, each electric vehicle's electricity consumption increases during the first and second stages, but each electric vehicle's electricity that is regenerated during the third stage decreases (see Fig. 3);

¹ In this paper, that the electricity consumption is negative denotes the electricity that the electric vehicle regenerates during its deceleration process.

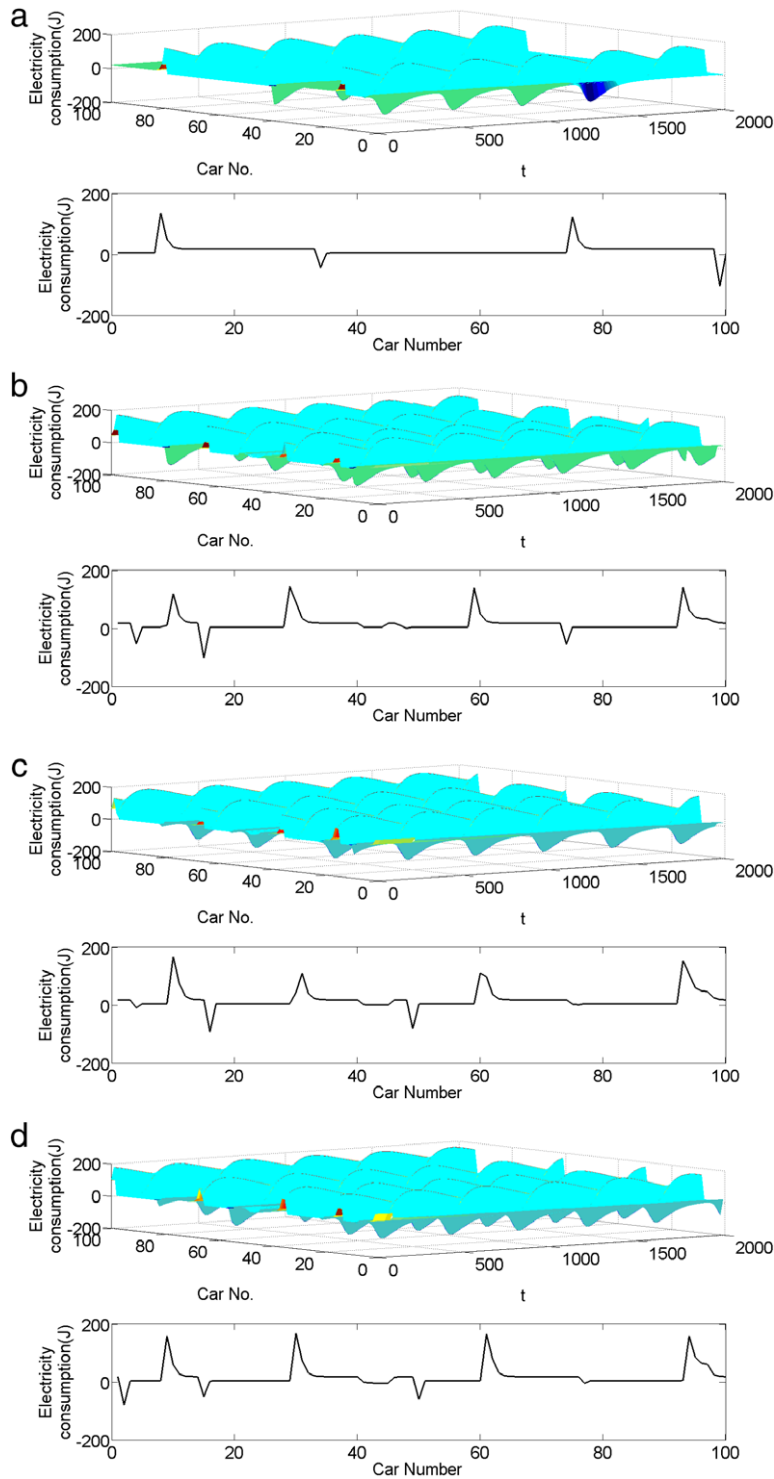


Fig. 5. The electric vehicle's electricity consumption after 3×10^5 time steps and the profile 3×10^5 th time step, where $L_{\text{uphill}} = 100$ m and the tilt angles in (a)–(d) are respectively 0° , 1° , 2° , 3° .

(3) with the increase of the downhill's tilt angle, each electric vehicle's electricity consumption decreases during the first and second stages, but each electric vehicle's electricity that is regenerated during the third stage increases (see Fig. 3).

To exhibit the relationship between the total electricity consumption and the road's slope, we calculate the 11 electric vehicles' total electricity consumption during the starting and braking processes (see Table 2). The data shown in Table 2

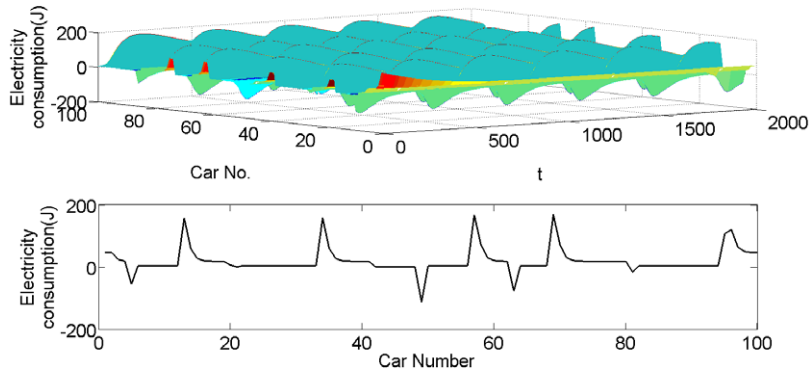


Fig. 6. The electric vehicle's electricity consumption after 3×10^5 time steps and the profile 3×10^5 th time step, where $L_{\text{uphill}} = 200$ m and the tilt angle is 2° .

Table 2

Total electricity consumption.

	Uphill			Downhill		
θ	1°	2°	3°	1°	2°	3°
Electricity consumption (10^6 J)	5.4988	7.0385	8.5913	2.4726	1.8938	1.3576

Table 3

Total electricity consumption during the period of 2000 time steps after 3×10^5 time steps.

θ	0°	1°	2°	3°
Electricity consumption (10^8 J)	4.6542	5.2862	5.3491	5.4355

Table 4

Total electricity consumption during the period of 2000 time steps after 3×10^5 time steps.

Uphill length (m)	100	200
Electricity consumption (10^8 J)	5.3491	5.8362

illustrate that the total electricity consumption increases with the increase of the road's tilt angle on uphill and decreases with the increase of the road's tilt angle on downhill.

Next, we explore the electric vehicle's electricity consumption when a small perturbation occurs on the ring road with different slope, where the initial conditions are defined as follows:

$$x_1(0) = 1 \text{ m}; \quad x_i(0) = (i-1)L/N, \quad \text{if } i \neq 1, \quad v_i(0) = V(\Delta x_i(0)), \quad (16)$$

where $N = 100$ is the total number of electric vehicles, $L = 1500$ m is the road's length. Here, we use the periodic boundary. Since we here consider the road's slope, we should give the following assumption:

- (1) the ring consists of four segments, i.e., uphill, flat road, downhill and flat road;
- (2) the tilt angles of the uphill and downhill are equal;
- (3) the lengths of the uphill and downhill are equal;
- (4) the lengths of the two flat roads are equal to $(L - L_{\text{uphill}} - L_{\text{downhill}})/2$, where L_{uphill} , L_{downhill} are respectively the lengths of uphill and downhill.

Other related parameters are the same those in Figs. 3 and 4. Thus, we can obtain each electric vehicle's electricity consumption under different tilt angles (see Figs. 5 and 6, Tables 3 and 4). From the two figures and tables, we can obtain the following results:

- (I) the electric vehicle's electricity consumption increases with θ on uphill and downhill (see Table 3);
- (II) when the tilt angle is 0° (i.e., the ring has no slope), the change of each electric vehicle's electricity consumption is slight; but when the ring has slope, the change of each electric vehicle's electricity consumption is relatively prominent and becomes more prominent with the increase of θ (see Fig. 5);
- (III) the total electricity consumption increases with L_{uphill} (see Fig. 6 and Table 4).

4. Conclusions

Researchers have explored electric vehicle's electricity consumption from different perspectives and obtained some important results, but little effort has been made to investigate the impacts of the road's slope on the electric vehicle's electricity consumption. In this paper, we use the theory of the electric vehicle's battery to study the effects of the road's slope on the electric vehicle's electricity consumption on uphill and downhill. The numerical results show that the electric vehicle's electricity consumption and the total electricity consumption increase with the road's tilt angle on uphill but decrease with the road's tilt angle on downhill. However, we do not use the empirical data to further calibrate the numerical results or the relationship between the electric vehicle's electricity consumption and the road's tilt angle. Therefore, we should in the future use empirical data or experiments to further explore this topic.

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